EE3032 Introduction to VLSI – Midterm

Fall 2022 – reconstructed original paper, 100 points

The complete original printed pages 2-1 and 2-2 are available. Text, numbering, marks and circuit connections are retained; handwritten annotations are not exam constraints.

1. (35 points) Answer the following questions briefly.

(a) List the process for transferring the mask pattern to the surface of a silicon region. (3 points)

(b) What is silicon on insulator process? Explain how it can reduce the leakage current. (4 points)

(c) List two methods for the oxidation in silicon. (2 points)

(d) List two features of static CMOS circuits. (2 points)

(e) List two components of CMOS circuit, which can result in static power. (2 points)

(f) In a logic gate, we usually implement the pull-up network with PMOS transistors and the pull-down network with NMOS transistors. Why? (2 points)

(g) One of modern CMOS process technologies provides high-k and metal gate transistor. Explain why the high-k material is used. (2 points)

(h) List two through-silicon via (TSV) process technologies. (2 points)

(i) In a CMOS gate, the transistors connected to the gate output suffer from the severe body effect. Why? (2 points)

(j) What is antenna effect? List two design methods for the fixing of antenna effect violation. (6 points)

(k) List three bonding technologies for 3D ICs. (3 points)

(l) The output capacitance of a CMOS logic gate in a CMOS circuit is comprised of three components. List the three components. (3 points)

(m) List two types of FinFET structures. (2 points)

2. (20 points) Fig. 1(a) shows the block diagram of an 8-to-1 multiplexer.

(a) Design the multiplexer using transmission gate. (10 points)

(b) What is the relationship of the multiplexer delay to the number of selection inputs ($s_0 s_1 s_2$). Explain your answer. (5 points)

(c) Assume that the circuit shown in Fig. 1(b) realizes the function $F = AB + \overline{B}C + D$. Complete the circuit. (5 points)

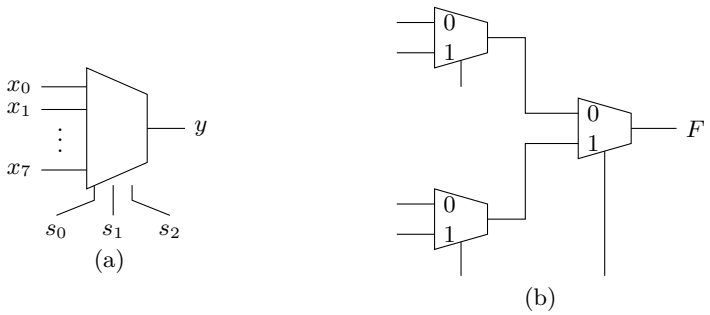


Figure 1: (a) 8-to-1 multiplexer. (b) Multiplexer-based circuit.

3. (10 points) Fig. 2(a) shows a CMOS circuit driving a large capacitance $C_L = 2.98\,\text{pF}$. For the used process, the input capacitance C_g of a minimum-size inverter (i.e., the size of NMOS and PMOS is W_{min}/L_{min}) is 1 fF. Assumptions: 1) the drain and wire parasitic capacitance can be neglected; 2) $e = 2.7$, $e^4 = 55$, $e^6 = 403$, and $e^8 = 2980$; 3) $\mu_n = 2\mu_p$; 4) the turn-on resistance of an NMOS transistor with W_{min}/L_{min} is R.

(a) Design a super buffer as shown in Fig. 2(b) to drive the large capacitance such that the output load of the NAND gate A is C_g and the delay T is minimized. (5 points)

(b) For the designed super buffer, if the size of NMOS transistors is fixed and we desire to tune the size of PMOS transistors such that the rise time and fall time of each inverter are equal, determine the size of PMOS transistors of the super buffer. Estimate the delay T in terms of R and C_g if the modified super buffer is used! (5 points)

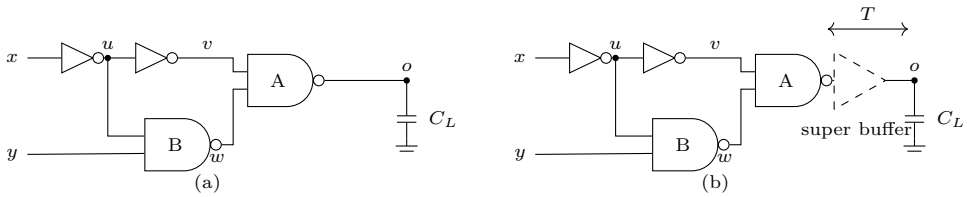


Figure 2: (a) A NAND gate driving a large capacitance. (b) A super buffer is inserted.

4. (15 points) Fig. 3(a) shows a CMOS logic gate. Assume that the threshold voltage of the NMOS transistor is 0.4 V; the turn on resistance of N1, N2, and N3 is R, $2R$, and R, respectively.

(a) If the initial voltage of the output V_{out} is 1.2 V and an input stimulus ($0 \rightarrow 1.2$ V) as shown in Fig. 3(b) is applied, how much voltage of the input stimulus (V_{in}) at x is? (9 points)

(b) Identify the operation region of the transistors N1, N2, and N3 from t_1 to t_2. (6 points)

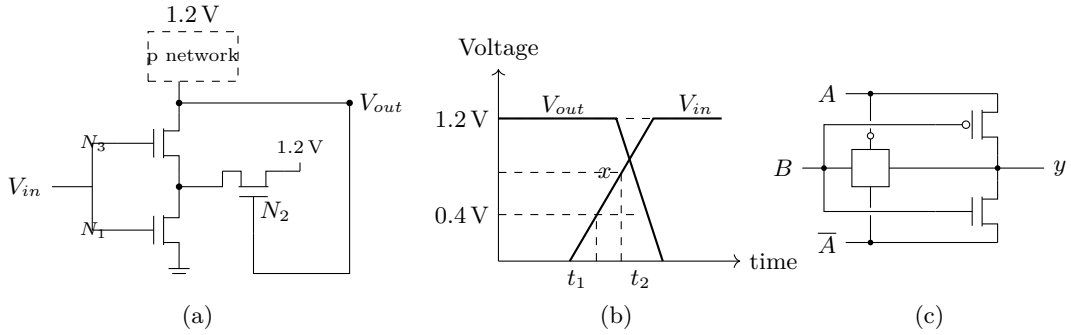


Figure 3: (a) A CMOS logic gate. (b) Waveforms of V_{in} and V_{out}. (c) A CMOS circuit.

5. (10 points) What function does the circuit shown in Fig. 3(c) realize? Does it have any weaknesses?

6. (10 points) Complete the following questions.

(a) Implement the following Boolean function: $F = X(Y + Z) + XW$ using CMOS logic. Draw the schematic diagram. (5 points)

(b) We have a CMOS logic gate realizing $F = \overline{X(Y + Z)}$ and the sizes of all the NMOS and PMOS transistors of the CMOS logic gate are W/L. Design an equivalent inverter for analyzing the rise time (t_r) and fall time (t_f) of the logic gate with input transitions $XYZ:\ 101 \rightarrow 100 \rightarrow 101$. (5 points)